1. Factorials

Factorial means multiplying a positive integer by all positive integers below it.

$$n! = n(n-1)(n-2)\cdots 1$$

Examples:

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$$

Important:

- $0! = 1$
- $1! = 1$
- $4! = 24$
- $6! = 720$

2. Fundamental Counting Principle

If one task can be performed in m ways and another in n ways, both tasks together can be performed in:

$$m \times n$$

Example:

You have:

- 3 shirts
- 2 pants

Total combinations:

$$3 \times 2 = 6$$

1. What is a permutation?

Permutation means arranging objects where **order matters**.

Example: We have 3 students: A, B, C. We need to select 2 students and arrange them in two positions.

Possible arrangements:

- AB
- BA
- AC
- CA
- BC
- CB

Total = 6.

2. Permutation formula

$$^{n}P_{r} = \frac{n!}{(n-r)!}$$

Where:

- n = Total objects
- r = Number of objects selected and arranged
- ! = Factorial

Example 1

How many ways can 3 students be arranged from 5 students?

$$^{5}P_{3} = \frac{5!}{(5-3)!}$$

$$= \frac{5!}{2!} = \frac{120}{2} = 60$$

Answer = **60**

Example 2: Arranging all objects

How many ways can 4 different books be arranged on a shelf?

$$4! = 4 \times 3 \times 2 \times 1$$

Answer = **24**

Remember: If all objects are arranged, use n!.

3. Important GATE concepts

Case 1: Repetition allowed

A 3-digit code using digits 1, 2, 3, 4, 5, where repetition is allowed:

$$5 \times 5 \times 5 = 125$$

Case 2: Repetition not allowed

$$5 \times 4 \times 3 = 60$$

Case 3: Arrangements with restrictions

Example: 4 people are arranged in a row. A and B must sit together.

Treat A and B as one block.

Now objects are:

(AB), C, D

Number of arrangements:

$$3! \times 2! = 12$$

We multiply by 2! because A and B can switch positions inside the block.

Q1. How many ways can 3 students be arranged from 6 students?

- A. 18
- B. 60
- C. 120
- D. 720

Q2. How many ways can 5 different books be arranged on a shelf?

- A. 25
- B. 60
- C. 120
- D. 240

Q3. How many 3-digit codes can be formed using digits 1,2,3,4 if repetition is allowed?

- A. 12
- B. 24
- C. 64
- D. 81

Q4. How many 3-digit codes can be formed using digits 1,2,3,4 without repetition?

- A. 12
- B. 24
- C. 36
- D. 64

Q5. Four people A, B, C, D are arranged in a row. A and B must sit together. How many arrangements?

- A. 6
- B. 8
- C. 12
- D. 24

1. Together problems (Block method)

Example: 5 people A, B, C, D, E are arranged in a row. A and B must sit together.

Step 1: Treat A and B as one block.

Now we have 4 objects:

(AB), C, D, E

Step 2: Arrange these 4 objects.

$$4! = 24$$

Step 3: A and B can switch places.

$$2! = 2$$

Final answer:

$$4! \times 2! = 48$$

2. Apart problems (Gap method)

Example: 4 boys and 3 girls are arranged in a row. No two girls should sit together.

Step 1: Arrange the boys.

$$4! = 24$$

Step 2: Create gaps around boys.

Step 3: Select 3 gaps for 3 girls and arrange them.

$$^5C_3 \times 3! = 10 \times 6 = 60$$

Step 4: Total:

$$4! \times 60 = \boxed{1440}$$

3. Repeated letters

Example: How many distinct arrangements of the word LEVEL?

Total letters = 5.

- L repeats 2 times.
- E repeats 2 times.
- V repeats once.

Formula:

$$\frac{5!}{2! \times 2!} = \boxed{30}$$

Why divide? Because identical letters swapping places doesn't create a new arrangement.

4. Circular permutations

When n distinct people sit around a circular table:

$$(n-1)!$$

Example: 5 people sitting around a circular table:

$$(5-1)! = 4! = \boxed{24}$$

Q1. Six people A, B, C, D, E, F are arranged in a row. A and B must always be together. How many arrangements are possible?

- A. 120
- B. 240
- C. 360
- D. 720

Q2. How many distinct arrangements of the word BANANA are possible?

- A. 30
- B. 60
- C. 90
- D. 120

Q3. Five boys and three girls are arranged in a row. No two girls should be adjacent. How many arrangements are possible?

- A. 14400
- B. 14400?
- C. 14400
- D. 7200

Q4. How many circular arrangements are possible for 6 distinct people around a round table?

- A. 120
- B. 720
- C. 24
- D. 60

Q5. How many distinct arrangements of the letters in MISSISSIPPI are possible?

- A. 34650
- B. 69300
- C. 17325
- D. 11550

Q1. Seven distinct people are arranged in a row. A must appear before B. How many arrangements are possible?

Q2. How many distinct arrangements of the word MATHEMATICS are possible?

Q3. Eight people sit around a circular table. A and B must sit together. How many circular arrangements are possible?

Q4. Five men and four women are arranged in a row. How many arrangements are possible if all four women sit together?

Q5. How many 4-digit numbers can be formed using digits 1,2,3,4,5,6 without repetition, if the number must be even?

Combinations (Selection)

New concept GATE DA 2027

First, remember the difference:

- Permutation = Arrangement (order matters).
- Combination = Selection (order doesn't matter).

1. What is Combination?

Suppose we have 4 students: A, B, C, D.

We need to select 2 students for a team.

Possible teams:

- AB
- AC
- AD
- BC
- BD
- CD

Total = 6 teams.

Notice that AB and BA represent the same team.

2. Combination formula

$$^nC_r = \frac{n!}{r!(n-r)!}$$

Where:

- n = Total available objects.
- r = Number of objects selected.

Example 1

Select 3 students from 6 students.

$$^6C_3 = \frac{6!}{3!(6-3)!}$$

$$= \frac{6 \times 5 \times 4}{3 \times 2 \times 1}$$

$$= \boxed{20}$$

3. Important properties

Property 1:

$$^nC_0 = {}^nC_n = 1$$

Property 2:

$$^nC_r = {}^nC_{n-r}$$

Example:

$$^8C_3 = {}^8C_5$$

Property 3:

$$^{n}C_{1} = n$$

Property 4:

$$^{n}P_{r} = {}^{n}C_{r} \times r!$$

4. GATE-level example

A group has 5 boys and 4 girls. Select 3 students such that exactly 2 are boys.

Step 1: Select 2 boys.

$$^{5}C_{2} = 10$$

Step 2: Select 1 girl.

$$^{4}C_{1} = 4$$

Step 3: Multiply.

$$10 \times 4 = \boxed{40}$$

Q1. Find 6C2.

- A. 12
- B. 15
- C. 30
- D. 36

Q2. Find 8C6.

- A. 15
- B. 20
- C. 28
- D. 56

Q3. How many ways can 3 students be selected from 7 students?

- A. 21
- B. 35
- C. 42
- D. 210

Q4. A group has 4 boys and 3 girls. How many teams of 2 boys and 1 girl can be selected?

- A. 12
- B. 18
- C. 24
- D. 36

Q5. Which statement is always true?

- A. nCr = nPr
- B. nCr = nC(n-r)
- C. nCr = r!
- D. nCr = n!

1. Selection with restrictions

Example: Select 3 students from 6 students. Student A must be selected.

Since A is already selected, we need to choose only 2 more students from the remaining 5.

$$\binom{5}{2} = 10$$

Shortcut: If one person must be selected, fix that person first, then select the remaining people.

2. Exactly k members

Example: A committee of 4 must contain exactly 2 women. There are 3 women and 5 men.

Choose 2 women and 2 men:

$$\binom{3}{2} \times \binom{5}{2} = 3 \times 10 = 30$$

Shortcut:

$$\text{Required selection} = \binom{\text{women}}{\text{required women}} \times \binom{\text{men}}{\text{required men}}$$

3. At least one

Example: Select 3 people from 4 men and 3 women. Find the number of groups containing at least one woman.

Use the complement method:

$$\text{At least one woman} = \text{Total groups} - \text{All-men groups}$$

$$= \binom{7}{3} - \binom{4}{3} = 35 - 4 = \boxed{31}$$

4. Two people must NOT be together

Example: Select 3 people from 6, including A and B. Find the number of groups where A and B are not both selected.

$$\text{Total groups} = \binom{6}{3} = 20$$

Groups containing both A and B:

$$\binom{4}{1} = 4$$

Therefore:

$$20 - 4 = \boxed{16}$$

Q1. A committee of 3 people is selected from 5 students. Student A must be included. How many committees are possible?

A) 4
B) 6
C) 8
D) 10

Q2. A committee of 4 is selected from 3 women and 4 men. Exactly 2 women must be included. How many committees are possible?

A) 12
B) 15
C) 18
D) 24

Q3. A group of 3 people is selected from 4 men and 3 women. How many groups contain at least one woman?

A) 25
B) 28
C) 31
D) 35

Q4. A group of 3 people is selected from 6 people: A, B, C, D, E, F. How many groups can be formed if A and B must NOT both be selected?

A) 12
B) 14
C) 16
D) 18

Q5. NAT: A committee of 5 is selected from 6 men and 4 women. Find the number of committees containing **at least 2 women**.

Concept 1: At least / at most using cases

Suppose we select 5 people from 6 men and 4 women, with at least 2 women.

Possible cases:

- Exactly 2 women and 3 men
- Exactly 3 women and 2 men
- Exactly 4 women and 1 man

Formula:

$$\sum_{k=2}^{4} \binom{4}{k} \binom{6}{5-k}$$

Concept 2: More women than men

For a committee of 5, more women than men means:

- 3 women, 2 men
- 4 women, 1 man
- 5 women, 0 men

Always identify possible cases before applying combinations.

Concept 3: Two people cannot be together

Use the complement method:

$$\text{Valid selections} = \text{Total selections} - \text{Selections containing both}$$

Concept 4: Multiple restrictions

When one person must be selected and two others cannot be together:

1. Fix the compulsory person.
2. Calculate all remaining selections.
3. Subtract the invalid selections.

Q1. MCQ – At least 2 women

A committee of 4 is selected from 5 men and 4 women. How many committees contain at least 2 women?

A) 76

B) 81

C) 85

D) 91

Q2. MCQ – More women than men

A committee of 5 is selected from 6 men and 4 women. How many committees have more women than men?

A) 60

B) 64

C) 66

D) 70

Q3. MCQ – Combined restrictions

There are 8 people: A, B, C, D, E, F, G and H.

Select a group of 3 people such that:

- C must be selected.
- A and B must not both be selected.

How many groups are possible?

A) 18

B) 19

C) 20

D) 21

Q4. MSQ – Selection properties

A group of 2 people is selected from 5 women and 4 men. Select all correct statements

A) Total possible groups = 36

B) Groups containing exactly one woman = 20

C) Groups containing at least one woman = 30

D) Groups containing at most one woman = 26

Q5. NAT – Advanced case analysis

A committee of 5 is selected from 7 men and 5 women.

Find the number of committees containing **at least 3 women**.

1. Inclusion–Exclusion principle

If A and B are two overlapping conditions:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

2. Exactly / at least / at most with multiple groups

For example, selecting 5 people with at least 2 women:

$$\sum_{k=2}^{5} \binom{W}{k} \binom{M}{5-k}$$

Remember to include only feasible cases.

3. Compulsory and forbidden members

- Must include A: fix A first.
- Must exclude B: remove B from the available group.
- A and B cannot both be selected: subtract groups containing both.

A committee of 4 is selected from 6 men and 5 women. The committee must contain at least 2 women and must include at least one man.

How many committees are possible?

There are 8 people: A, B, C, D, E, F, G, H.

Select 4 people such that:

- A must be selected.
- B must not be selected.

How many selections are possible?

From 10 students, a team of 3 is selected. Two particular students A and B must not both be selected. How many valid teams are possible?

A committee of 6 is selected from 7 men and 5 women. Find the number of committees containing at least 2 women.

There are 9 people: A, B, C, D, E, F, G, H, I.

Select a group of 4 such that:

- A must be included.
- B and C cannot both be selected.

Find the number of valid groups.

A committee of 4 is selected from 5 men and 4 women. How many committees contain exactly 2 women?

There are 10 students. A must be selected and B must not be selected. How many ways can a team of 5 be formed?

From 8 people, a team of 3 must be selected. Person C is compulsory, and A and B cannot both be selected. How many teams are possible?

A committee of 5 is selected from 7 men and 5 women. Find the number of committees containing at least 3 women.

From 12 people, a team of 4 is selected. A must be included, B must be excluded, and C and D cannot both be selected.

Find the number of possible teams.

A committee of 5 is selected from 6 men and 5 women. Find the number of committees containing more women than men.

From 8 people, A and B are two specified people. A committee of 4 must contain either both A and B or neither A nor B.

Find the number of possible committees.

A committee of 5 members is to be selected from 7 men and 5 women. The committee must contain at least 2 women and at least 2 men.

How many committees are possible?

There are 10 students A, B, C, D, E, F, G, H, I, J. A must be selected, B must not be selected, and C and D cannot both be selected.

A committee of 4 students is formed. How many committees are possible?

There are 8 people A, B, C, D, E, F, G, H. A committee of 5 must be selected.

Conditions:

- A must be included.
- B must be excluded.
- C and D cannot both be selected.

Find the total number of valid committees.

A committee of 5 is selected from 6 men and 5 women.

Conditions:

- At least 2 women.
- At least 1 man.
- Two particular women, A and B, cannot both be selected.

How many valid committees are possible?

There are 10 people. A must be selected, B must be excluded, and C and D cannot both be selected.

A committee of 5 is formed. How many valid committees are possible?

There are 9 people A, B, C, D, E, F, G, H, I.

A committee of 5 must be selected under these conditions:

- A and B must both be selected.
- C must not be selected.
- D and E cannot both be selected.

Find the number of valid committees.

In how many ways can 6 distinct people A, B, C, D, E, F sit in a row if A and B must sit together?

A committee of 4 is selected from 6 men and 5 women. How many committees contain at least 2 women?

In how many ways can 7 distinct people sit around a circular table if A and B must sit together?

How many distinct arrangements can be formed using all letters of the word **BANANA**?

How many 4-digit numbers can be formed using digits 1, 2, 3, 4, 5, 6 without repetition if the number must be even?

A committee of 5 is selected from 7 men and 5 women. The committee must contain at least 3 women.

Find the total number of possible committees.

MSQ – Combination identities

Which of the following statements are always true for valid integer values of n, r?

A) $\binom{n}{r} = \binom{n}{n-r}$

B) $\binom{n}{r} + \binom{n}{r-1} = \binom{n+1}{r}$

C) $\binom{n}{0} = 0$

D) $\binom{n}{1} = n$

Which statements are always true for valid values of n, r?

A) $\binom{n}{2} = \frac{n(n-1)}{2}$

B) $\binom{n}{r} = \binom{n}{r+1}$

C) $\binom{n}{0} = 1$

D) $\binom{n}{r} + \binom{n}{r-1} = \binom{n+1}{r}$

A team of 4 is selected from 6 men and 5 women. Which statements are correct?

A) Number of teams with exactly 2 women = 150
B) Number of teams with at least 1 woman = 320
C) Number of teams with at least 2 women = 265
D) Number of teams with at most 2 women = 285

For integers $n \geq 2$, which identities are always correct?

A) $\binom{n}{0} + \binom{n}{1} = n + 1$

B) $\binom{n}{r} = \binom{n}{n-r}$

C) $\binom{n}{r} + \binom{n}{r-1} = \binom{n+1}{r}$

D) $\binom{n}{1} = \binom{n}{n-1} = n - 1$

A committee of 4 is formed from 6 men and 5 women. It must contain at least 2 women and at least 1 man.

Which statements are correct?

A) Exactly 2 women and 2 men can be selected in 150 ways.

B) Exactly 3 women and 1 man can be selected in 60 ways.

C) Total valid committees = 210.

D) Selecting 4 women is allowed.

There are 9 students A, B, C, D, E, F, G, H, I. A must be selected, B must be excluded, and C and D cannot both be selected. A team of 4 is formed.

Which statements are correct?

A) Before applying the C/D restriction, there are 35 teams.

B) Teams containing both C and D = 5.

C) Valid teams = 30.

D) Valid teams = 25.

A team of 5 is selected from 7 men and 5 women.

Which statements are correct?

A) Teams with exactly 3 women = 350

B) Teams with at least 3 women = 371

C) Teams with at most 2 women = 371

D) Teams with more women than men = 371

For $0 \leq r \leq n$, which statements are always true?

A) $\binom{n}{r} = \binom{n}{n-r}$

B) $\binom{n}{r} + \binom{n}{r+1} = \binom{n+1}{r+1}$

C) $\binom{n}{2} = \frac{n(n-1)}{2}$

D) $\binom{n}{0} + \binom{n}{1} = n^2$

A committee of 3 is selected from 5 people A, B, C, D, E.

Which statements are correct?

A) Total committees = 10

B) Committees containing A = 6

C) Committees excluding A = 4

D) Committees containing both A and B = 3

Five distinct people are arranged in a row.

Which statements are correct?

A) Total arrangements = 120

B) Arrangements with A first = 24

C) Arrangements with A and B together = 48

D) Arrangements with A before B = 60

A team of 4 is selected from 5 men and 4 women.

Which statements are correct?

A) Teams with exactly 2 women = 60

B) Teams with at least 1 woman = 121

C) Teams with at least 3 women = 21

D) Teams with no women = 5

Six distinct people A, B, C, D, E, F are arranged in a row. A and B must be together.

Which statements are correct?

A) Treating A and B as one block gives 5 objects.

B) Internal arrangements of A and B = 2.

C) Total valid arrangements = 240.

D) Total valid arrangements = 120.